Fiscal Year:	FY 2017	**Task Last Updated:**	FY 08/03/2017
PI Name:	Hammond, Timothy M.D.		
Project Title:	Yeast Colony Survival in Microgravity Depends on Ammonia Mediated Metabolic Adaptation and Cell Differentiation -Flight		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	None		
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Project Type:	Flight	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	08/01/2013	**End Date:**	01/31/2018
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
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Flight Program:	ISS		
Flight Assignment:	Rapid Turn Around Flight ISS NOTE: End date changed to 1/31/2018 per F. Hernandez/ARC (Ed., 8/21/17 NOTE: Extended to 7/31/2018 per F. Hernandez/ARC (Ed., 7/31/17) NOTE: End date changed to 7/31/2017 per NSSC information (Ed., 7/25/16) NOTE: End date is 7/31/2016 (originally 7/31/2015) per NSSC information (Ed., 3/27/14)		
Key Personnel Changes/Previous PI:	May 2015 report: Added Drs. Corey Nislow and Guri Giaever as Co-Is.		
COI Name (Institution):	Birdsall, Holly M.D., Ph.D. (Baylor College of Medicine) Nislow, Corey Ph.D. (University of British Columbia) Giaever, Guri Ph.D. (University of British Columbia)		

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Task Description:	Yeast chemical genomics is a powerful tool that is being applied to drug discovery. This approach allows the study of compound-target relationships in the context of a living cell with a systems biology view that includes both the cell target and the interdependent networks that respond to chemical stress. The modulation of yeast colonies, especially the redox potential, allows study of different drug pathways. This application investigates microbial metabolic adaptation and cell differentiation to the multiple stimuli encountered in space flight environments, using the yeast Saccharomyces cerevisiae as a model system. We have shown that microgravity activates a unique set of genes in yeast. Our previous work has largely defined yeast cellular stress defense response to microgravity, but stress defense responses are only involved in the survival of yeast in liquid cultures. It has recently been found that survival of plate-based yeast colonies is dependent on signaling mediated through ammonia which drives the multicellular yeast colony to differentiate into two populations, L or lower cells and U or upper cells. These two populations have different metabolic pathways and establish a cycle of nutrients that allows the U cells to continue to proliferate at the leading edge. Without ammonia signaling, the entire colony dies. Lack of convection in space is likely to dramatically impact signaling by ammonia, a volatile gas. We postulate that the flow of nutrient and ammonia signals will differ in microgravity where there is little to no convection. We hypothesize that yeast colony survival in microgravity depends on ammonia mediated metabolic adaptation and cell differentiation rather than on stress defense. To test our hypothesis, we will identify the mechanisms mediating colony stratification into U and L cells under multiple gravitational environments: microgravity emulated by Rotating Wall Vessel and Random Positioning Device, static one g cultures, and ultimately microgravity of space flight. Our experimental model will compare liquid and solid phase yeast cultures and make use of available mutant strains of yeast with deletion of the enzymes involved in ammonia signaling and the connected alternative metabolism activation, enzymes involved in stress responses, and the Msn4 promoter and SSA4 gene that we have previously identified as mediating response to microgravity. Relevant metabolic pathways will be studied using green fluorescent protein-linked reporter genes that can be analyzed by confocal microscopy of intact colonies, and flow cytometry of cells divided into L and U populations by sucrose density centrifugation. There are similarities between multicellular yeast colonies and tumors of clusters of mammalian cells in tissue culture. Thus insights into the effect of microgravity on cell to cell signaling may have far-reaching applications. Microgravity could provide a physiologically responsive test bed for drug pathway analysis at diverse redox states.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Yeast chemical genomics is a powerful tool that is being applied to drug discovery. This approach allows the study of compound target relationships in the context of a living cell with a systems biology view that includes both the cell target and the interdependent networks that respond to chemical stress. It has recently been found that survival of plate-based yeast colonies is dependent on signaling mediated through ammonia which drives the multicellular yeast colony to differentiate into two populations, L or lower cells and U or upper cells. Lack of convection in space is likely to dramatically impact signaling by ammonia, a volatile gas. There are similarities between multicellular yeast colonies and tumors of clusters of mammalian cells in tissue culture. Thus insights into the effect of microgravity on cell to cell signaling may have far-reaching applications. Microgravity could provide a physiologically responsive test bed for drug pathway analysis at diverse redox states.
Task Progress:	Baker’s yeast is a model eukaryote used for centuries to prepare bread and beer, is a powerful tool with biomedical applications including cell-to-cell interactions such as colony differentiation and stratification as well as cell autonomous behavior such as cell survival, aging, and cancer drug action. Yeast have broad genetic homology to human cells and are simple to genetically manipulate. Reactive oxygen system pathways play a central role as mediators of all of these effects, and assays for reactive oxygen system pathways have been optimized and validated for use in yeast studies. When grown as giant (greater than 10 mm in diameter) colonies on nutrient agar, yeast colony stratification is indelibly linked to gravity-driven convection, as ammonia production and the gravity driven convection of produced ammonia mediate giant yeast colony differentiation. Growth and differentiation of giant yeast colonies is a month-long, iterative cycle of cell differentiation, production of signaling molecules, and stratification of different cell phenotypes. Rotating culture devices randomize the direction of convection vectors, and have been a popular model to mimic the minimal convection induced by microgravity during spaceflight. Reduced convection disrupts ammonia signaling, altering the redox changes of the yeast cells and gene expression kinetics during growth of giant yeast colonies. Our lab has previously studied the responses of yeast to real (flight-based) and simulated microgravity. We have used a deletion strategy to demonstrate that mitochondrial and ribosomal redox gene pathways play a predominant role in the responses of yeast colonies to microgravity. This genetic modulation of yeast colonies in real and simulated microgravity is dependent, at least in part, on both ammonia and the shear stress promoter Sfp1, but not the stress-responsive Msn4 promoter. The random positioning machine (RPM), referred to as the 3-D clinostat, is a microgravity simulator based on the principle of 'gravity-vector-averaging', which can generate simulated gravity levels from 0g to 1g. There is limited data comparing the RPM and simpler 2-D rotation, and controversy as to whether the forces they induce are distinct. The current study is the first to make a direct comparison of disrupting ammonia convection in giant yeast colonies by the use of rotating cultures on the ground, and in the true microgravity of space, using static cultures as a baseline. This experimental design allowed us to address the following questions; which features of giant yeast colony differentiation are shared or different in rotating cultures and the microgravity of space? Are there gene expression or redox space-based changes not seen on the ground? Are there quantitative differences between spaceflight and rotating cultures? Does the RPM offer advantages over simple 2-D rotation? Does media composition (e.g., nutrient rich vs nutrient poor) alter phenotypic responses in spaceflight compared to rotating ground cultures? Definitions of these effects are important to facilitate translation of yeast genetics to pharmacology, tumorogeniesis, and commercial applications.

	The importance of these studies is to determine if spaceflight or the RPM can expand the dynamic range and sensitivity of studies of redox changes and allow experimental modulation of reactive oxygen status by physical rather than pharmacological means, to provide a drug-free genetically intact system to define the effects of reactive oxygen status on drug metabolism pathways. In this study, we used physical factors including colony rotation, and microgravity to modify ammonia convection and shear stress. We observed differences in the responses of both ammonia dependent and stress response dependent pathways. We found that the effects of random positioning is distinct from clinorotation. Furthermore both true and modeled microgravity exacerbated cellular redox responses These changes were largely shear response dependent but each model had a unique response signature as measured by shear stress genes, and the promotor set which regulates them. These physical techniques permitted a graded manipulation of both convection and ammonia signaling, while avoiding secondary effects of gene deletion, and are primed to substantially contribute to our understanding of mechanisms of drug action, cell aging, and colony differentiation. The data has been submitted for publication. Further analysis of yeast deletion pools by next generation sequencing is ongoing. Sequencing libraries have been prepared, and sequencing performed. Gene Set Enrichment Analysis (GSEA) and Cytoscope/ClueGo visualization is rapidly being completed. We are sequencing deeper to confirm if observed ribosomal transcriptional changes are robust.
Bibliography Type:	Description: (Last Updated: 03/31/2026)
Articles in Peer-reviewed Journals	Hammond TG, Stodieck L, Koenig P, Hammond JS, Gunter MA, Allen PL, Birdsall HH. "Effects of microgravity and clinorotation on the virulence of Klebsiella, Streptococcus, Proteus, and Pseudomonas." Gravit Space Res. 2016 Jul;4(1):39-50. http://gravitationalandspacebiology.org/index.php/journal/article/view/719 , Jul-2016